The Augmented Synthetic Historical Control Method

Introduction

Causal inference in economic and policy research often relies on comparing a treated set of units to a control or donor pool. The synthetic control method (SCM) of Abadie et al. [2010] is one of the most widely used approaches in panel data econometrics, second only to difference-in-differences. The idea of SC is straightforward: the untreated potential outcomes of a treated unit are approximated by a weighted average of donor units. Difference-in-differences relies on the assumption of parallel pre-intervention trends [Baker et al., 2025].

In many settings, however, no suitable donor units are available—for example, during global events such as pandemics or financial crises, or when no valid donor days exist. To address this, Chen et al. [2024] proposed the synthetic historical control method (SHC). Rather than relying on external donors, SHC constructs a donor pool from the treated unit's own pre-treatment history, splitting the pre-treatment period into overlapping segments. A quadratic program then estimates weights over these segments to predict counterfactual outcomes. Like Abadie et al. [2010, 2015], SHC restricts weights to the simplex. This guarantees interpretability but can be too restrictive when noise or complex dynamics make exact pre-treatment fit unattainable.

This paper introduces the Augmented Synthetic Historical Control method (ASHC), which extends SHC in two key ways. First, ASHC relaxes the simplex constraint to an affine constraint, requiring only that weights sum to one. This permits negative weights, allowing controlled extrapolation beyond the convex hull of observed segments in exchange for better fit. Second, ASHC adds a quadratic proximity penalty that discourages large deviations from the SHC solution, in the spirit of the augmented SCM [Ben-Michael et al., 2021]. This encourages the estimator to remain close to SHC unless extrapolation is needed. ASHC retains the stabilizing Gram penalty from Chen et al. [2024], which regularizes low-variance directions in the donor matrix. Together, these three components yield a new estimator that generalizes SHC to settings where exact pre-treatment fit is otherwise impossible.

The paper makes three contributions. Methodologically, it introduces the ASHC estimator, which generalizes SHC by permitting limited, penalized extrapolation while maintaining the stabilizing Gram penalty. Theoretically, it derives finite-sample error bounds that decompose ASHC prediction error into four interpretable parts: the SHC residual error, a regularization-controlled deviation term, smoothness-induced bias, and noise. It further establishes asymptotic consistency and a convergence rate of $O_p(m^{-H/(2H+3)})$, which approaches the near-parametric rate $O_p(m^{-1/2})$ as the smoothing order H increases. Here H denotes the smoothing order (the number of Taylor expansion terms retained), while h_m denotes the kernel bandwidth, with the optimal choice balancing bias $O(h_m^H)$ and variance $O_p((mh_m)^{-1/2})$. When the pre-treatment length m is large (e.g. $m = 120$) and H grows, the rate becomes practically close to the parametric $O_p(m^{-1/2})$, so ASHC achieves efficient performance in realistic sample sizes. Practically, it provides simulation and empirical evidence showing that ASHC yields stable and accurate counterfactuals in settings where SHC alone fails to achieve adequate pre-treatment fit.

The discussion begins with a careful review of SHC, its assumptions, and its implementation, before introducing the ASHC framework. We then turn to the theoretical analysis, establishing finite-sample error bounds and asymptotic properties. The empirical performance of ASHC is explored through calibrated simulations, followed by an application that illustrates its usefulness in practice. Proofs are collected in the appendix, and all computations are carried out using Python's mlsynth library (https://mlsynth.readthedocs.io/en/latest/).

References

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